SMM921 coursework: summer 2026

Base your analysis on the tools, measures and examples that we have covered in the lectures. If you introduce any measures or tools that were not covered in the lectures, you must describe them fully and provide citations to their sources.

Students may use GenAI in a supporting role in this assessment (e.g., in proofreading or in testing and refining code), but must remain the author of the text submitted.

Throughout this report, use 12pt font, 1.25 line spacing and standard margin size. Figures and tables should be embedded in the report and all count towards the 20 page limit. Add citations where appropriate and useful and include a list of references at the end of the document, in Harvard style.

Part 1: Liquidity analysis: [this part is worth 50% of the final mark]

Extract the data in the archive file `SMM921_trading_data_2026.csv.zip` that is on the Moodle page. This file contains intra-day data, sampled at a 1 minute frequency, for around 20 stocks for 3 months at the beginning of 2026. There is an observation recorded at the end of every minute and the data columns comprise a stock identifier, a date and time variable, the price of the most recent trade, shares traded in the last one-minute interval, the number of trades in the interval, the bid, the ask and shares displayed at the bid and ask at the end of the interval. [Note: if, for a particular minute and a particular stock, there is no data in the last price, shares traded and number of trades columns, you should interpret this as meaning that there were no trades in that minute.]

Randomly select **three** of these stocks and perform the analysis below on those three. (Your final sample of three stocks should not simply be the first three stocks in the file or the last three stocks in the file and you might want to pick stocks with a good variety of market caps.)

Create midquotes, midquote returns, spread data and depth data for each minute. Spreads should be computed as the ratio of the difference between the ask and the bid prices to the midquote and expressed in basis points. Depth should be computed at each point in time as the average of the GBP value of the shares available at the best bid and ask (note that this implies that one needs to compute products of shares available and the price per share). Then filter all observations before 8:15 and after 16:25 in each day. Also filter any observation where the spread is negative. You should also check for outliers and, if you identify any that you believe should be removed, then remove them and report your reasons for removal in the report that you submit.

Take the perspective of a quantitative portfolio manager and trader. Create a short report containing no more than 10 pages which describes, interprets and compares the average liquidity of your chosen stocks. Also show how liquidity changes across the trading day (on average) for each of your stocks and describe the implications of this. Finally, for each stock investigate, using regression and correlation analysis, how daily liquidity is related to daily volatility (defining daily midquote return volatility as the average value of all the absolute minutely returns within a given day). Last, give indications of any shortcomings in your analysis, what that might mean for anyone using these results to trade the stocks and describe how you might remedy them.

Part 2: portfolio analysis: [this part is worth 50% of the final mark]

Take the data in the file `SMM921_PF_DATA_2026.xlsx` that is on the Moodle page. This file contains 20 years of monthly data on a cross-section country equity index levels. (These levels are adjusted for dividends so that percentage changes computed from them are total returns.) Create monthly returns. Create a 'world stock market return' for each month equal to the average return across the countries.

Write a short report of no more than 10 pages from the perspective of a quantitative portfolio manager which should focus on the following issues.

- Based on (annualised) mean returns, return standard deviations and on Sharpe ratios, describe and compare how these countries have performed. Also, using the world return as the 'market portfolio', compute the systematic risk of each country, then describe and compare your findings. (Note: I do not want an analysis of each individual country, but some discussion and interpretation of the top few and bottom few performers would be useful.)
- Your manager is interested in testing a basic momentum signal and wants you to do the quantitative work. Define momentum to be the total return for a country over the twelve months leading up to the current observation excluding the most recent month. Define 5 momentum-sorted portfolios, each containing roughly 1/5th of the assets, rebalanced every month and compute their returns. Also create a HML portfolio return equal to, in each month, the return on the highest momentum portfolio minus the return on the lowest momentum portfolio. In your analysis, give plots, summary statistics, performance measures, risk measures and interpretations of each of these for the five portfolio returns and for the HML returns.
- Now your manager wants you to perform a genuine portfolio optimization based on the same momentum signal. Start at the 61st month in the sample. Create alphas for the optimization from your momentum signals, using the methods covered in the lectures to transform the momentum signals to alphas. Assume an information coefficient of 0.02 and residual return volatility equal to the cross-sectional mean of the volatilities of the country-level return series over the most recent 60 data points. Create a covariance matrix of returns using the most recent 60 data points. Finally, assume a risk-aversion coefficient of 4. With these inputs create mean-variance optimal weights that change every month and then create the monthly returns that one earns from holding those weights over the subsequent month. In your report, provide relevant plots, summary statistics, performance measures, risk measures and interpretations for the optimal portfolio return series.
- Your manager is unconvinced by your previous optimized portfolio construction analysis and wants you to use a more robust covariance matrix construction. She

suggests that, at each observation, instead of using the sample covariance matrix over the most recent 60 data points, you modify that matrix to leave the variances unchanged but you base all the covariances on the mean correlation implied by that sample covariance matrix. Redo all of your previous optimization analysis based on this method, present results and compare with those that you derived earlier. What do you infer and what is your final recommendation from the analysis?

How I will judge your analysis.

Precision in calculations and the level of sophistication in your computational and statistical analysis is vital. **Equally** important is clarity of explanation of the **implications** of your findings from a portfolio management perspective. A passing mark will require you to perform strongly in both of these aspects.

Guidance on use of generative AI in this assignment

Students may use GenAI in a supporting role in this assessment (e.g., in proofreading or testing and refining code), but must remain the author of the text submitted.

Attach a GenAI acknowledgement (≤150 words) as a final page of your submitted assignment (this does not count towards the page limit). Submissions without this acknowledgement will be treated as incomplete until it is received and will be returned for completion. The acknowledgement should take the following form.

GenAI Assistive Use Acknowledgement

Tool(s) used (name + version/URL): ____

Purpose(s) (tick): ☐ proofreading ☐ clarity/tone ☐ translation ☐ code testing ☐ brainstorming/outlining ☐ other: ________

What I changed/kept based on the GenAI output (1–2 sentences): ____________________
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Verification I performed (tick): ☐ checked facts/calculations ☐ verified references ☐ ran code/tests ☐ compared to sources ☐ plagiarism check (non-AI)

Limits respected (1–2 sentences on what I did not use AI for, per brief): ________________
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Declaration: I am the author of this work. AI was used only for the assistive purposes stated. No personal/sensitive data or third-party confidential material was uploaded to non-approved tools. I accept responsibility for accuracy, originality, and proper citation.

Name/ID & date: ____